Harbin Institute of Technology

Master's Thesis Proposal

Title: Benchmarking and Improving LLMs for Reasoning in Urdu using Chain-of-Thought and Retrieval-Augmented Generation

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Major: Computer Science and Technology

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Date: September 11, 2025

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1. Source, Objectives, and Significance

1.1. Source of the Research Project

The Shenzhen Science and Technology Program is the source of this project and fits into the current attempts to improve natural language processing of low-resource languages. Large Language Models (LLM) have shown considerable reasoning capabilities on high-resource languages like English and Chinese, but it has not had much exploration on the reasoning capabilities of its models regarding Pakistani languages, and more so, Urdu. The previous Urdu NLP research has concentrated on classification such as sentiment analysis, hate speech detection, and fake news detection (Harris et al., 2024 [1]; Tahir et al., 2024 [2]), and machine reading comprehension (Kazi and Khoja, 2025 [3]). Nonetheless, the initiatives focus on superficial knowledge as opposed to a multi-step thinking process.

The gap demonstrates the necessity to have an Urdu reasoning benchmark. In the absence of such a resource, one cannot assess and refine the capabilities of LLMs to process arithmetic problems, commonsense reasoning, and multi-hop factual questions in Urdu. The urgent solution is to deal with this issue since Urdu is the national language in Pakistan with more than 70 million native speakers and over 200 million speakers in communication. In this way, the project will develop an Urdu Reasoning Benchmark (URBench) to test LLM reasoning in Urdu in a systematic way and examine lightweight improvement techniques like Chain-of-Thought (Wei et al., 2022 [4]), self-consistency (Wang et al., 2022 [5]), or Retrieval-Augmented Generation (Lewis et al., 2020 [6]).

1.2. Objectives and Significance

Problem. Large Language models (LLMs) are highly resourceful, reasoning in high resource languages, including English and Chinese with

large datasets and benchmarks. Nevertheless, their performance in Pakistani languages particularly the Urdu is still poor and mostly untested. Available Urdu NLP systems are limited to surface-level tasks such as sentiment analysis, classification, and machine reading comprehension and do not provide a benchmark specific to testing multi-step reasoning. Such scarcity complicates the measurement of whether in Urdu LLMs are able to reason in arithmetic, reason logically, or reason about commonsense

The absence of such benchmarks means that Urdu LLM can easily give shallow answers, hallucinations and inconsistent reasoning, thus restricting its educational utility and practical use. This divide limits the development of such spheres as education, governance, and access where trustworthy Urdu reasoning systems may make a tangible change. Hence, the lack of systematized datasets and assessment systems to benchmark and enhance the LLM reasoning in Urdu is fundamentally the problem.

Objective. Build URBench (Urdu Reasoning Benchmark), perform baseline tests (direct prompting vs CoT/self-consistency) and apply RAG to a small Urdu knowledge base to decrease hallucination in multi-hop questions. While keeping the scope manageable firstly I'll focus only on text data and keeping the room to expand my work to multi-modal with time-to-time.

Significance. The work offers (a) the first dedicated Urdu reasoning benchmark, including arithmetic and commonsense, multi-hop factual QA, instruction-following, and procedural reasoning (b) reproducible evaluation scripts/metrics; and (c) practical improvement recipes, not involving training new models, useful in academia and industry alike with interest in Urdu or similar low-resource environments.

2. Current Research Status and Analysis

2.1. Status at Home and Abroad

In Large Language Models (LLMs), reasoning has been examined at scale through benchmarks, including GSM8K (Cobbe et al., 2021 [7]), StrategyQA (Geva et al., 2021 [8]) and MMLU (Hendrycks et al., 2020 [9]) that reasoning arithmetic, commonsense and domain-specific reasoning. Recent developments indicate that inference-time prompting strategies, including Chain-of-Thought (Wei et al., 2022 [4]) and self-consistency (Wang et al., 2022 [5]) can considerably enhance reasoning capacity, and Retrieval-Augmented Generation (Lewis et al., 2020 [6]) has become a common method of justifying factual answers. These studies emphasize that the reason behind the ability of the LLM thinking can be analytically quantified and enhanced, yet studies have focused primarily on English and other well-resourced languages.

On the contrary, the studies on the Pakistani languages, in particular, Urdu are few. The current resources relate to classification and reading comprehension, including the UQA corpus of Urdu question answering (Arif et al., 2024 [10]) and UQuAD+ of machine reading comprehension with semantic-aware evaluation (Kazi and Khoja, 2025 [3]). Sentiment analysis (Harris et al., 2024 [1]), hate speech, and fake news detection (Tahir et al., 2024) are covered by other studies; however, none of them provides the benchmark of multi-step reasoning. There is a huge gap here: despite the ability to generate fluent Urdu text, the reasoning capacity of LLMs in Urdu has not been measured or improved in a systematic way. This vacuum must be filled so that strong Urdu applications emerge in the education, governance and access to information.

More recently, evaluation studies like the CHiPSAL-1.3 shared task (Tahir et al., 2025 [2]) have shown that even state-of-the-art multilingual models such as GPT-3.5, Bloomz, and LLaMA combine inconsistently on Urdu work, and that semantic and reasoning-oriented evaluations in particular

are weak. These results support the need to create task-specific Urdu benchmarks that go beyond the classification into reasoning. With a focus on basing on international approaches to CoT and RAG and basing it on the realities of the Urdu script, morphology, and cultural background, this project will hopefully address a research gap that is quite obvious and create a starting point of further research in Urdu reasoning.

2.2. Literature Review and Analysis

As the existing literature on reasoning methods indicates, there are apparent possibilities of adjusting them to Urdu. Chain-of-Thought prompting has also been shown to be very effective with mathematical and logical reasoning based on encouraging step-by-step solutions, and self-consistency enhances reliability by sampling many reasoning paths and choosing the one that is the most consistent answer (Wei et al., 2022 [4]; Wang et al., 2022 [5]). Equally, Retrieval-Augmented Generation (RAG) focuses on basing model outputs on external knowledge to encourage reduced hallucination and is thus especially useful in knowledge-based tasks (Lewis et al., 2020 [6]). Taken together, these techniques constitute a viable set of tools that can be used in improving reasoning performance without the need to retrain the same at high costs.

In comparison, the current Urdu NLP tools are still confined to the question answering (UQA, Arif et al., 2024 [10]; UQuAD+, Kazi and Khoja, 2025 [3]), sentiment and hate speech (ISE-Hate, Harris et al., 2024) and news categorization tasks. These contributions show that it is possible to construct Urdu datasets but not multi-step reasoning. There is no systematic evaluation of Urdu in the same way that GSM8K (Cobbe et al., 2021 [7]) and StrategyQA (Geva et al., 2021 [8]) would be used in English.

This discussion shows that the research gap is evident: Urdu does not have a standard by which the reasoning ability is tested in a variety of tasks. The proposed URBench dataset is meant to address this gap by scaling reasoning-style tasks to Urdu and by complementing them with

inference-time enhancements like CoT, self-consistency, and RAG. This makes sure that the project is not only in line with the international progress but also serves a local concern that is burning.

2.3. Case Studies on Urdu Reasoning Challenges

Case Study 1: Translation vs. Reasoning.

The UQA dataset (Arif et al., 2024 [10]) demonstrated that it is possible to get a large-scale QA pairs in the Urdu language using translation-based corpora. Nonetheless, assessments indicate that models tend to acquire pattern matching and not actual reasoning because the tasks are similar to span extraction (such as SQuAD). It means that models have the capability of retrieving the answers but cannot relate more than two facts or reason out in-between steps.

Case Study 2: Evaluation Reliability in Urdu

Semantic match (SM) was proposed in the UQuAD+ dataset (Kazi & Khoja, 2025 [3]) to solve the exact-match problems. As an illustration, two semantically identical responses (3 vs. 3) can be graded as incorrect using conventional measures. This emphasizes the fact that the evaluation of Urdu is not reliable unless semantic-sensitive measures are taken and this directly reflects on reasoning benchmarks.

Case Study 3: Broader Benchmarking Gaps

The CLIPSAL shared task (Tahir et al., 2025 [10]) was a metric of multiple Urdu NLP challenges including classification and summarization where up to 17 tasks were benchmarked by the participating LLMs. As far as the coverage was concerned, however, the coverage was comprehensive, there were no reasoning-based reasoning tasks (Multi-hop QA, commonsense, procedural reasoning), though. The absence of this points to an urgent need of certain Urdu standards of reasoning such as URBench.

Case Study 4: Commonsense Reasoning Gaps

Most Urdu datasets are sentiment classification and hate speech, and news classification, but there are no commonsense reasoning datasets (such as English CommonsenseQA or StrategyQA) in Urdu. As an example, the question of whether Ali puts milk in the freezer and what will happen after two hours. demands daily reasoning and not recall of facts. The existing Urdu standards do not challenge this kind of logic and a critical gap is present whether any commonsense is supported by the LLMs in low-resource languages.

Case Study 5: Code-Switching and Roman-Urdu

The excessive use of the code-switching (mixing Urdu with English) and Roman-Urdu scripts on the social media is a particular challenge of Urdu reasoning. As an example, “Kal class hai ya holiday”? combines both English and Urdu. The existing samples are almost confined to pure Urdu writing, which disregards the actual linguistic usage. This weakness indicates that LLMs that have been trained or assessed using the current corpora might not work well in veritable conversational reasoning scenarios.

Case Study 6: Multimodal Reasoning in Urdu

The latest LLMs such as GPT-4V and Gemini emphasize the need to use multimodal reasoning (text plus image or text plus speech). Urdu, however, does not have multimodal datasets in, say, reasoning over charts or diagrams, or speech. As an example, understanding a bus schedule in Urdu or thinking over an Urdu captioned picture is still uncharted. This exclusion constrains the range of Urdu AI applications, especially regarding accessibility (e.g. education to users of the visually impaired kind).

3. Main Research Content and Plan

3.1. Research Content

This research will focus on URBench (Urdu Reasoning Benchmark) and test LLMs on this dataset and conduct some experimentation on how to improve the data in practice. The data will consist of approximately 600-800 reasoning problems in five categories: (i) arithmetic word problems, (ii) commonsense reasoning, (iii) multi-hop factual question answering, (iv) instruction-following with constraints and (v) procedural/step-by-step reasoning. All questions are to be presented in a table with the following fields: id, category, question_ur, answer_ur, answer_norm and source, which must be reproducible and easy to evaluate. To create this dataset, I will combine translated and adapted English benchmarks (GSM8K and StrategyQA) with native Urdu items written and screened by bilingual speakers, and ensure that they are culturally and linguistically appropriate. Besides, in order to reflect the actual application of language in Pakistan, the code-switched items within the context of Urdu-English combination will appear in the data (about 10-15%).

Table: Proposed Schema for URBench (Urdu Reasoning Benchmark)

Field Name	**Description**	**Example (Urdu/English mix for clarity)**
ID	Unique Identifier for each item	URB-001
Category	Task Type (Arithmetic, Common-sense, Multi-hop, instruction, procedural)	Arithmetic
Question_UR	Full Question in Urdu	علی کے پاس 3 سیب ہیں، وہ 2 مزید خریدتا ہے، کل کتنے ہیں؟
Answer_UR	Gold standard answer	"پانچ"

	in Urdu	
Answer_norm	Normalized answer (Western digits/simplified text for evaluation)	5
Rationale	Optional step-by-step reasoning English/Urdu mix used for COT baselines)	“ 3 + 2 = 5”
Source	Origin of the item (translated GSM8K, native-authored, etc)	GSM8K-translated
Split	Dataset Partition (Train, Dev, Test)	Train

3.2. Evaluation Metrics

Various measures of model performance will be used according to the type of task. Precision and pass@1 will be used to confirm that the first answer of a model matches the gold standard in arithmetic and constraint-following tasks (exact match (EM) accuracy and pass). To count self-consistency (best-of-n sampling), I will also report consistency@n, by verifying whether one of several sampled reasoning paths can produce the correct answer. In short factual or span questions, EM and F1 scores will be derived when the Urdu script variations such as diacritics and digits have been normalized. In multi-hop factual tasks, an LLM-as-a-judge format will be employed, where answers are scored as correct or incorrect based on a rubric, and other checks will be inserted to determine whether the model makes valid citations in responses based on RAG or not. ROUGE scores (ROUGE-1, ROUGE-2, ROUGE-L) will be provided in summarization-like answers or more extended answers. Taken together, these measures will help to establish a fair and transparent evaluation of various reasoning tasks

3.3. Models and Baselines

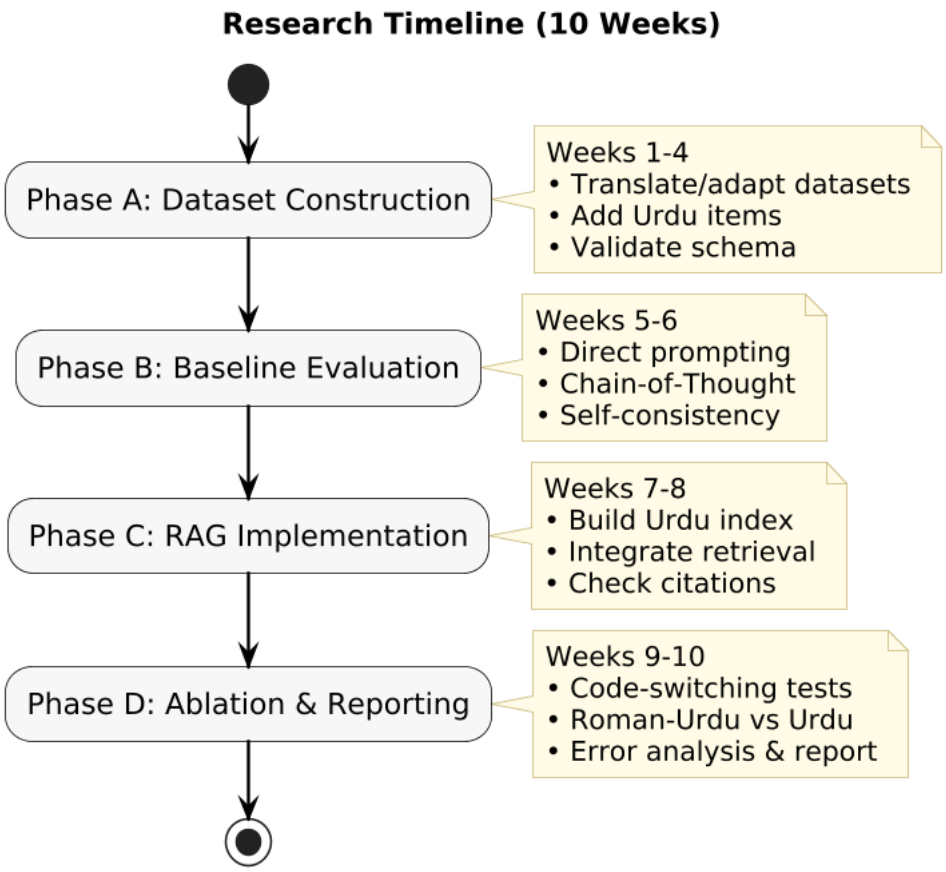
Both frontier LLMs (e.g., GPT-4, Claude) available as API will be considered, as well as open-source multilingual LLMs (e.g., LLaMA 3, Qwen, Mistral) which can be deployed to any cloud platform (e.g., Google Colab or Hugging Face). The baseline conditions will include direct prompting in Urdu, Chain-of-Thought prompting and self-consistency of multiple reasoning pathways. In case of open models, when there are only limited resources, only smaller parameter sizes will be assessed, but where possible results of larger versions will be also reported. The dual evaluation approach enables a comparison of closed, high performing models and open, re-producible models

3.4. Improvement Methods

We will test three lightweight improvement methods on the top of the baselines. The first will be Chain-of-Thought (CoT) prompting that will require the model to produce intermediate steps ("آئیے قدم بہ قدم سوچتے ہیں / let's think step by step") before giving a final answer. Second, to evaluate the notion of self-consistency, the sampling of the reasoning paths (n=5) and choosing the most common final answer or a logically consistent one will be used, which has proven to improve the reasoning reliability significantly. Third, the architecture of Retrieval-Augmented Generation (RAG) pipeline will be used based on a small Urdu corpus (Urdu Wikipedia and selected news articles). This pipeline will be based on multilingual embeddings and FAISS as a retrieval system, and the retrieved passages will be suggested in the prompt to anchor the model answers and minimize hallucination. A light multimodal slice can be added as an option extension with simple text+image reasoning tasks (e.g., solving math problems with diagrams) or text+Speech inputs (using Urdu TTS/ASR), but this component is small and does not require scaling

3.5. Research Plan and Methodology

The study will be written in four stages. Phase A (Dataset Construction, Weeks 1-4): translate, adapt, write Urdu reasoning items; debug with bilingual reviewers; write data schema and normalization programs. Phase B (Baseline Evaluation, Weeks 5-6): enable direct prompting, CoT and self-consistency baselines on all categories; display results in tables and charts. Phase C (RAG Implementation, Weeks 7-8): construct a retrieval index over Urdu Wikipedia/news and combine with LLM prompts; test the effect on multi-hop factual tasks, such as hallucination and citation tasks. Phase D (Ablation and Reporting, Weeks 9-10): carry out ablation experiments on code-switched items, Roman-Urdu vs Urdu vs pivot-to-English prompts; interpret error data; draft results, figures, write up. Python, Jupyter Notebook, Hugging Frameworks Datasets and Transformers, FAISS or LangChain to retrieve information, and spreadsheets to annotate and analyze the datasets will be used.



4. Expected Objectives and Contributions

4.1. Deliverables

There are four primary deliverables that the research will generate. The first one is the URBench dataset, a mini-benchmark of approximately 600-800 Urdu reasoning problems. Activities will involve arithmetic (e.g.,"علی کے پاس 3 سیب وہ، ہیں 2 مزید خریدتا ہے، کل کتنے ہیں؟"), commonsense (e.g., "ہو رہی ہے کیا تو کیچھ تری ضرورت ہے؟ بہار شاگر"), procedural, multi-hop factual and instruction-following reasoning. Other such datasets in English, like GSM8K (Cobbe et al., 2021), arXiv link, have already contributed substantially to research on reasoning; URBench hopes to do the same in Urdu.

The second deliverable is a baseline assessment report, where (i) direct prompting in Urdu, (ii) Chain-of-Thought (CoT) prompting (Wei et al., 2022) [4], and (iii) self-consistency (Wang et al., 2022) [5] are compared in a systematic way. These techniques are also known to enhance the quality of reasoning in English; their use in Urdu will attempt to determine how well they can be transferred to a low-resource language.

The third deliverable is a Retrieval-Augmented Generation (RAG) pipeline based on Urdu Wikipedia and news corpora, which was implemented with FAISS-based retrieval. RAG was proposed by Lewis et al. (2020) [11] and has become a common way of ensuring that LLMs receive factual information as their grounding. This pipeline is to be tested in the Urdu in factual and multi-hop reasoning and hallucination is most problematic here.

The fourth product is a reusable bundle of scripts and codes, such as data building scripts, assessment measures, experiment templates. This was inspired by open resources, such as Hugging Face Datasets [12], (to make URBench transparent, reusable and extensible to future researchers.

4.2. Novelty

The work fills a definite gap directly: there is no standard used to assess multi-step reasoning in Urdu. The available Urdu materials are centered on QA or comprehension, rather than reasoning- e.g. the UQA corpus in translation of SQuAD2.0 (Arif et al., 2024 [10]) and manually annotated UQuAD + a semantic-match measure (Kazi and Khoja, 2025 [3]).More general assessments of Urdu NLP tasks have performance of classification and MRC but do not separate reasoning out (Tahir et al., 2024 [2]). URBench one of its novelty is that it is a specialized benchmark evaluating arithmetic, commonsense, and multi-hop reasoning in Urdu.

Mathemetically, the project is the first to systematically adopt inference-time reasoning techniques into Urdu, i.e., Chain-of-Thought prompting (Wei et al., 2022 [4]).and self-consistency (Wang et al., 2022 [5]) as well as Retrieval-Augmented Generation (RAG) to factually ground (Lewis et al., 2020 [6]). The techniques have previously been supported by high-resource languages; this is the first systematic test in Urdu, to indicate what is transferred and what needs to be adapted.

In practice, the novelty is both a resource-efficient reproducible recipe: all the improvements are made at inference time (no expensive fine-tuning), and a small, well-documented benchmark and open evaluation scripts. This reduces the barrier to both academic and industry teams that operate in a low-resource environment and provides a standard, repeatable method of measuring and refining Urdu reasoning.

4.3. Academic and Practical Impact

This project has immense academic value due to the fact that it brings into the scene a benchmark not present in the Urdu NLP. GSM8K (Cobbe et al., 2021, [7]) and MMLU (Hendrycks et al., 2020, [9]) are some benchmarks

that have been instrumental in the development of reasoning research in the English language. Equally, URBench can serve as a benchmark of Urdu reasoning, enabling researchers to gauge progress on a systematic scale and even to rank models on an equal footing. The reproducible evaluation scripts, schema of the dataset, and baseline reports will be used as a learning and research tool by students and faculty in AI programs in Pakistan and around the world. This not only builds a local capacity, but also makes Urdu a constituent in the global discourse on LLM reasoning.

URBench has real-world implications in more than just one area of application. Education AI An Urdu reasoning standard can directly facilitate the creation of AI tutoring systems that can solve math word problems, logic puzzles, or factual questions in Urdu, and make digital education more accessible to millions of students. The RAG pipeline will enable Urdu question-answering systems used in journalism and media to give citation-based responses that are fact-checked, which is a major challenge in fighting misinformation and fake news in Pakistan. An example would be having a RAG-enhanced LLM respond to inquiries about a political event by quoting the Urdu news story rather than hallucinating a response.

Urdu reasoning is of immediate use in government and other works of state. Numerous citizens speak mostly Urdu and the implementation of AI systems able to follow instructions like “اس قانون کا 3 نکات میں خلاصہ کریں” (“Summarize this law in three points”) would increase availability and transparency. Likewise, a chatbot serving a public hotline might be much more helpful provided they are capable of reasoning based on a multi-step guide as opposed to just responding to factual questions.

The other real effect is a reduction in barriers to academia and industry. Since this project involves inference-time methods but not a massive retraining, solutions are cheap and can be adopted even by smaller organizations, startups, or individual scientists. Making the code and

dataset scripts open-source would mean URBench is not a donation but a code that can be built upon in the future alongside the community.

Lastly, there is the long-term effect that is both regional and multilingual. This project provides a prototype of how the reasoning benchmarks and inference-time techniques can be applied to Urdu, and, based on this prototype, other Pakistani languages, including Punjabi, Sindhi, and Pashto, can be extended. These languages also have no standards of reasoning, and maybe URBench can encourage such work, with the increased scholarly worth and social impact. This way, the study is not just an Urdu study but also a first of its kind in the history of inclusive AI in South Asia.

5. Completed Work and Schedule

5.1. Completed Work (Related Studies and Foundations)

I have made a number of steps before I undertook the proposed project in order to have a strong foundation. To start with, I executed a small analysis of the leading UNE sources like UQA (Arif et al., 2024 [10]) and UQuAD+ (Kazi and Khoja, 2025 [3]) to know the way datasets are constructed and tested in low-resource settings. This was by investigating their translation processes, annotation schemes and metrics, especially Semantic Match (SM). This analysis has provided insight into the trade-offs between large-scale translation-based corpora, which have a risk of having cultural or language mismatches, and expensive but high-quality manual annotation. Such lessons have a direct impact on my design decisions of URBench on balancing between scalability and reliability.

Second, I learned significant reasoning standards and algorithms in high-resource languages, such as GSM8K, StrategyQA, and MMLU, and inference-time algorithms, such as Chain-of-Thought prompting (Wei et al.,

2022 [4]), self-consistency (Wang et al.,2022 [5]), and Retrieval-Augmented Generation (Lewis et al., 2020 [6]). To develop a concept of practical familiarity, I conducted small exploratory experimentation with prompting in Colab with open-access models, which enabled me to have a first-hand experience of how reasoning styles vary between direct prompting and CoT. These experiments helped me to reinforce my conception of how such procedures can be modified to low-resource settings without retraining.

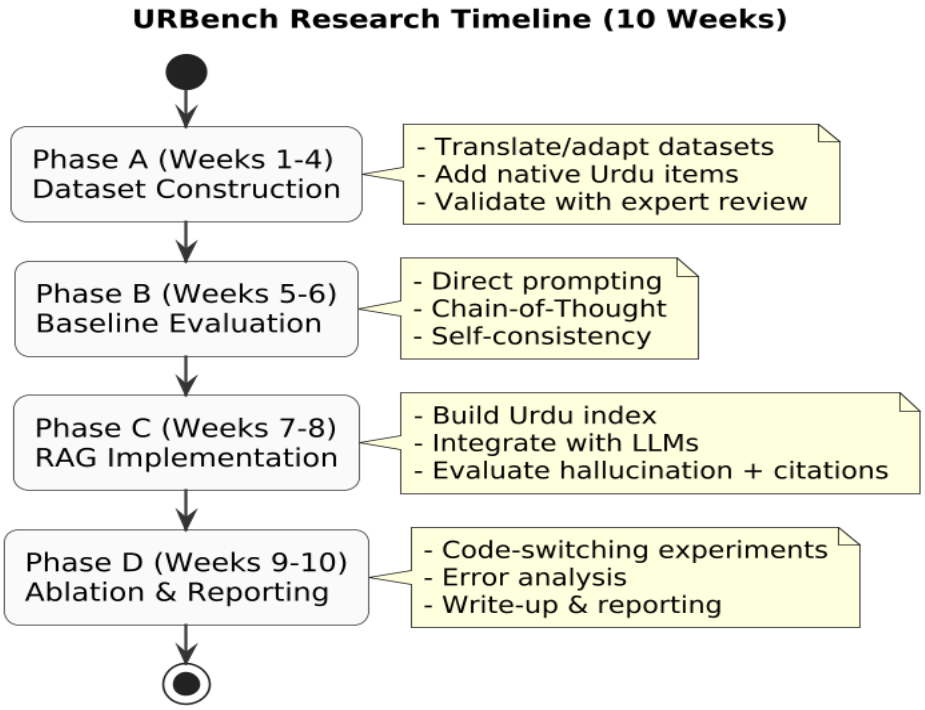
Third, I started searching Urdu corpus like the Urdu Wikipedia and online news items to determine how they would be effective in the RAG pipelines. I did small-scale coverage checks on such areas of coverage as science and policy and observed where the information is unavailable. This was accompanied by trying Hugging Face data and model APIs, testing the preprocessor, such as digit and diacritic normalization, and experimenting with embeddings using FAISS to gain a clearer idea of the demands of dense retrieval. The experiments performed at an early stage provided me with a practical experience of the technical tools that will be the foundation of my project

Fourth, I conducted a pilot translation exercise, translating a small set of English reasoning problems (e.g. math word problems in GSM8K) into Urdu. This mini-test proved that the process of translation was possible and that it was difficult to maintain logical consistency between languages. It offered a good starting point to design culturally and linguistically adequate items of reasoning in Urdu

Lastly, I was able to combine these steps into a systematic research plan that has definite stages: dataset construction, baseline evaluation, RAG integration and ablation studies. I also conceptualized this workflow using timeline diagrams that were used to map every preparatory activity to tangible deliverables. This preparation shows that I do not start at zero, but I have some background of theoretical and practical work with tools,

datasets and models used to run URBench.

5.2. Research Schedule



URBench Research Timeline

The research will be carried out in four phases over a period of 10 weeks:

- **Phase A (Weeks 1–4): Dataset Construction**
  - Translate and adapt reasoning datasets (e.g., GSM8K, StrategyQA) into Urdu.
  - Add a smaller set of culturally relevant, native Urdu items.
  - Validate data quality with expert review.
- **Phase B (Weeks 5–6): Baseline Evaluation**
  - Run direct prompting, Chain-of-Thought prompting, and self-consistency on URBench.
  - Compare results with multilingual baselines.

- **Phase C (Weeks 7–8): RAG Implementation**
  - Build an Urdu Wikipedia/news index.
  - Integrate with LLMs for multi-hop factual QA.
  - Measure hallucination reduction and citation accuracy.
- **Phase D (Weeks 9–10): Ablation and Reporting**
  - Explore Roman-Urdu/code-switching subsets.
  - Perform error analysis.
  - Write and prepare the thesis proposal/report.

6. Research Conditions and Funding Assurance

6.1. Existing Research Environment

The study will be conducted in the Harbin Institute of Technology, Shenzhen (HITSZ) where computational research that supports natural language processing is conducted. The college has regular computing facilities and study facilities. The programmable language will be Python and the software-side component of the experiments will be recorded in Jupyter Notebooks in order to ensure the repeatability of the experiments. The preparation of datasets, tokenization and evaluation of models will be centralized in libraries like Hugging Face Transformers and Datasets (Wolf et al 2020) [6]. In order to implement the Retrieval-Augmented Generation (RAG) pipeline, the following tools will be engaged: FAISS (Johnson et al 2019) [13] and LangChain (LangChain documentation) [14] to create retrieval indexes and combine the knowledge retrieved into prompts. Free research platforms like Google Colab (Google Research 2018) [15], Hugging Face Spaces [16], and API calls will be used to support experimentation on open-source models like LLaMA and Qwen, and access GPT-family models and Claude. Also, the human resources is provided by the research group of my supervisor that has bilingual peers,

who may help to check the translations of the dataset and assure the use of cultural information. A combination of these resources provides a good base upon which the proposed research is to be conducted.

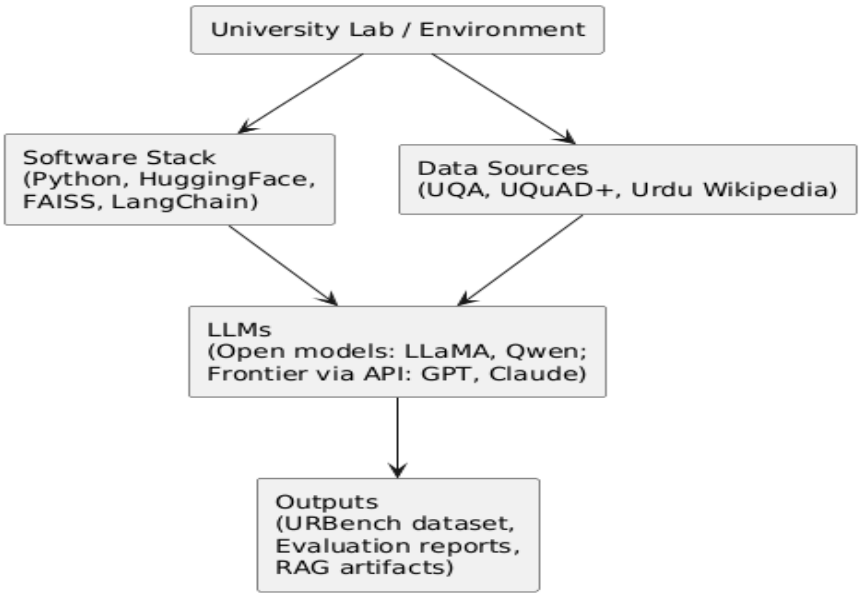
A simple table listing what resources we have vs. what we need

6.2. Computational and Data Resources

One of the benefits of this project is that it does not need to train large-scale models, which usually need GPU clusters and large compute. Rather, it targets inference-time enhancements, which include Chain-of-Thought prompting, self-consistency, and grounding retrieval. This option will make the project affordable and viable with the resources available. To construct the dataset, the available English benchmark such as GSM8K is used (Cobbe et al., 2021) [7] and StrategyQA (Geva et al., 2021) [8] will be translated and converted to Urdu. They are open data that are commonly used in the study of reasoning and are therefore viable both legally and ethically. The generated URBench dataset will be stored in a structured format (e.g. JSONL) and stored on GitHub or Hugging Face Datasets Hub with support of reproducibility. Cloud storage will not be great as the dataset will probably be small (around 600-800 items). To assess the project, it will be based on multilingual and Urdu-enabled models provided in Hugging Face Model Hub (Hugging Face Hub) [17].So are OpenAI GPT-4 (OpenAI, 2023) [18] and OpenAI GPT-4.or Anthropic Claude (Anthropic, 2023) [19] These resources have enough coverage to test both open-source and frontier models without special hardware.

Resource Type	Examples / Tools	Purpose in Project	Availability
Software	Python, Hugging Face, FAISS, LangChain	Dataset construction, evaluation, RAG pipeline	Available (open-source)
Hardware	Google Colab,	Running open	Available

	Hugging Face Spaces, university PCs	models, experiments	(free/low-cost)
APIs	GPT-4 (OpenAI), Claude (Anthropic)	Frontier LLM evaluation	Limited (pay-as-you-go credits)
Data	Urdu Wikipedia, UQA, UQuAD+	Retrieval and evaluation corpora	Available (public)
Human Resources	Bilingual peers, supervisor feedback	Validation of translations and quality	Available



Workflow for URBench

6.3. Funding Assurance and Feasibility

The budget of the project is relatively low as compared to finetuning or training projects. Key areas of investment will include: (i) API credits of GPT-4, Claude, or other models, which can be acquired on a pay-as-you-use basis at a relatively low cost; (ii) small annotation or review stipend, which bilingual reviewers will purchase to verify a sample

of the dataset translations and cultural suitability, as is the case in other datasets like UQuAD (Kazi & Khoja, 2025) [3]; and (iii) cloud hosting or storage, which can be handled through free or low-cost providers including GitHub or Hug As the project is developed with lightweight improvements such as Chain-of-Thought prompting and RAG, the costs do not exceed departmental or student level funds. Furthermore, open-source platforms like Hugging Face, FAISS, and LangChain make sure that the software stack is open-source and free to use under permissive licenses. On the whole, the project has a high feasibility since the computational and financial needs are small, and the required infrastructure is already available at HIT Shenzhen.

7. Anticipated Difficulties and Solutions

7.1. Anticipated Difficulties

Although the proposed research is designed to be feasible within the available resources, several difficulties may arise during its execution. Identifying these challenges in advance is important, as it allows for preventive planning and reduces risks during implementation.

The former issue involves the quality of datasets and their translation. Because a section of URBench will be translated from existing English benchmarks, like GSM8K and StrategyQA, there is a possibility that automatic translation will result in ambiguous or culturally unrelated Urdu text. Indicatively, a math problem that talks of dollars and miles can be translated directly to Urdu which will not sound natural as compared to “روپے روا کلومیٹر.” These inconsistencies may confuse the annotators as well as the models. Likewise, the actual translation of idioms or culturally-specific references can lose its meaning. This may make the evaluation results less reliable when models are disadvantaged due to language clarity instead of logic.

One of the challenges is reliability of evaluation. The Urdu writing system has variants of writing (e.g. optional diacritics, Western and Eastern number systems, high usage of English code-switching), which may lead to string matching that is strictly stricter than that of model performance. Similar metrics such as the Semantic Match (SM) of UQuAD+ (Kazi and Khoja, 2025 [3]) are thus better in place than exact matching.

The second is a constraint on compute and API costs. Chain-of-Thought and self-consistency methods are efficient inference-time techniques that nevertheless still need repeated query access to frontier models, like GPT-4 or Claude, potentially limited by credits or access to GPUs when using open models like LLaMA or Qwen.

The third problem is retrieval quality in RAG. Although the Urdu Wikipedia and news can be utilized, they are less wide-ranged than English and in other cases, multilingual embedding models do not coincide with the Urdu semantic, which poor retrieval has been shown. Even though FAISS (Johnson et al., 2019 [13]) is efficient with dense search, its morphologically rich Urdu performance has not been studied thoroughly.

7.2. Proposed Solutions

I will address the problem of dataset quality by using automatic translation and manual inspection of a part of the items by bilingual speakers to guarantee the linguistic and cultural accuracy. For example, reviewers can adapt “miles” into “کلومیٹر” or substitute non-local contexts with local contexts. The methodology is based on the validation practices used in UQuAD+, where bilingual supervision resulted in better reliability of the data sets (Kazi and Khoja, 2025). Selective review provides quality at affordable rates whereas full manual annotation is resource-consuming.

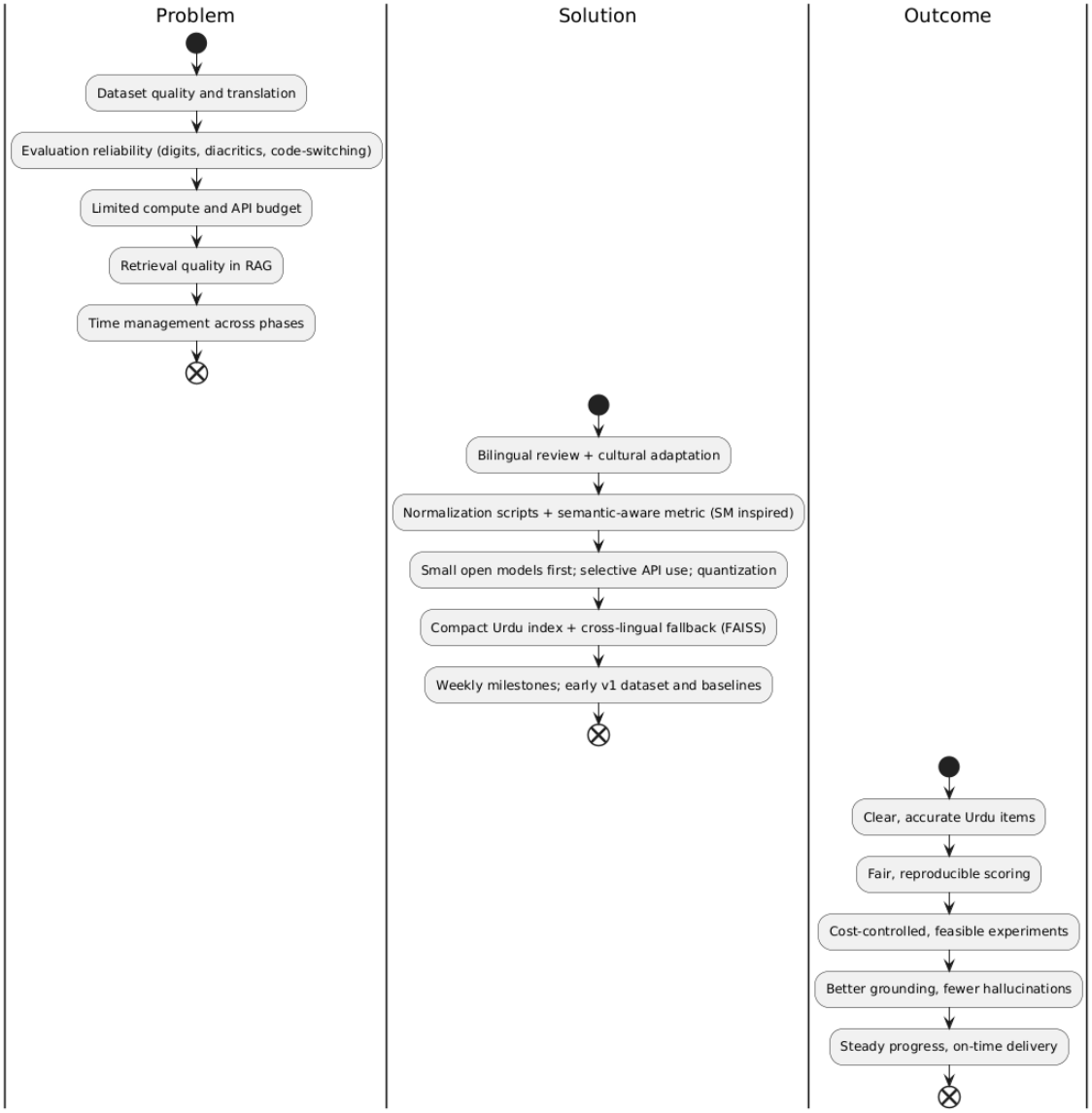
To address the issues of evaluation, I will create normalization scripts to standardize Urdu-specific variations such as turning all numerals into a single representation, removing optional diacritics, and normalizing similar spelling forms. Besides, I will implement semantic-sensitive scoring based on the SM measure of UQuAD+, so that answers like “۳” and “3” or “دو گھنٹے” and “2 hours” are said to be semantically equivalent. This makes the evaluation based on the ability to reason and not on the orthography.

To overcome the constraint of compute, I will use a model evaluation strategy in stages. The initial validation of datasets and metrics will be made using smaller open models (e.g. quantized 7B LLaMA versions) with low memory needs. Frontier model such as GPT-4 and Claude will be set aside until last tests, where just a small but representative part of the data will be checked to keep the API costs down. This is an effective and complete method of discrimination. CoT method will be injected earlier into the evaluation pipeline to capture the maximum insights without retraining costs (Wei et al., 2022) [4].

In RAG related issues, I will begin with a small but good retrieval index with the help of Urdu Wikipedia and big news sources. To reduce coverage gaps I will experiment with cross-lingual retrieval where queries can access relevant passages in English which can be translated to Urdu and loaded into the model. Such a backup plan will be a compromise between Urdu authenticity and proper coverage. FAISS will also be the reference indexing solution given its scalability and density retrieval capabilities (Johnson et al., 2019) [13].

Finally, in order to reduce the risks associated with time management, I will also implement a weekly milestones system. Indicatively, at the end of Week 4, a rough version of URBench will be ready and can be used to run baseline tests; at the end of Week 6, we will have baseline results; and at the end of Week 10, RAG pipeline and reporting will be complete. Each

of the milestones allows one to see the progress clearly and ensures that the delay of one step will not completely halt other steps.



Problem–Solution–Outcome Activity Diagram

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